1 Applications of Matrices and Determinants



Brahmagupta

(c.598 AD(CE) - c.668 AD(CE))

Introduction

In our daily life, We use Matrices for taking seismic surveys. They are used for plotting graphs, statistics and also to do Scientific studies in almost different fields. Matrices are used in representing the real world data like the traits of people's population, habits etc..

Determinants have wonderful algebraic properties and occupied their proud place in linear algebra, because of their role in higher level algebraic thinking.

Brahmagupta (born c.598 AD(CE) died c.668 AD(CE)) was an Indian Mathematician and astronomer. He was the first to give rules to compute with zero. His contribution in Matrix is called as Brahmagupta Matrix.

$$B(x\,,\,y) = \begin{pmatrix} x & y \\ \pm ty & \pm x \end{pmatrix}.$$

Learning Objectives

On Completion of this chapter , the students are able to understand

- the concept of rank of a matrix.
- elementary transformations and equivalent matrices.
- echelon form of a matrix.
- the rank of the matrix.
- testing the consistency of a non- homogeneous linear equations.
- applications of linear equations
- the concept of Cramer's rule to solve non- homogenous linear equations.
- forecasting the succeeding state when the initial market share is given.

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XII Std - Business Maths & Stat EM Chapter 1.indd 1 2/28/2020 5:27:11 PM